

Why Open Source Large Language Models are Popular

Open source large language models mark a pivotal moment in the evolution of generative AI. By lowering barriers to entry and fostering collaborative innovation, these models are enabling a broader spectrum of organisations to benefit from AI.



Generative artificial intelligence has ushered in a new era of innovation, transforming the way machines understand and generate human language. At the heart of this revolution are large language models (LLMs), which have become indispensable for tasks ranging from text generation to complex reasoning. While commercial LLMs have garnered significant attention, the emergence of open source alternatives has democratised access to advanced AI capabilities. These models, developed collaboratively by global communities, empower organisations and individuals alike to harness, adapt, and refine generative AI for a multitude of use cases.

Popular open source large language models

Several open source LLMs have risen to prominence, each offering unique features, architectures, and advantages. The Llama series, developed by Meta, exemplifies efficient and scalable language modelling, attracting

widespread adoption in research and industry. EleutherAI's GPT-Neo and GPT-J initiatives have successfully replicated and extended the capabilities of earlier transformer models, providing robust alternatives with permissive licences. Falcon, another noteworthy entrant, focuses on optimising inference speed and resource efficiency, making it suitable for deployment in cost-sensitive environments. Models such as BLOOM and MPT have further expanded the ecosystem, emphasising multilingual support and modular architectures. The collaborative development and transparent release of these models have fuelled rapid innovation and adaptation across the AI landscape.

The benefits: The adoption of open source LLMs confers several significant benefits. Foremost is the flexibility to customise and fine-tune the models according to specific organisational needs, which is often constrained in commercial alternatives. Cost efficiency is another major advantage, as the absence of licensing

fees lowers barriers for startups, research institutions, and enterprises. Transparency is inherent to open source projects, allowing practitioners to scrutinise model architectures, training data, and underlying assumptions. This openness cultivates trust and facilitates the identification and mitigation of biases. Additionally, the vibrant global community supporting open source LLMs accelerates knowledge sharing, troubleshooting, and collective improvement, ensuring that the technology evolves rapidly and responsibly.

Common use cases of genAI

Open source generative AI models find application in a diverse array of industries, each benefiting from tailored solutions. In healthcare, LLMs assist in medical documentation, summarisation of patient records, and even support clinical decision-making with evidence-based suggestions. The finance sector leverages these models for automating customer service, generating financial